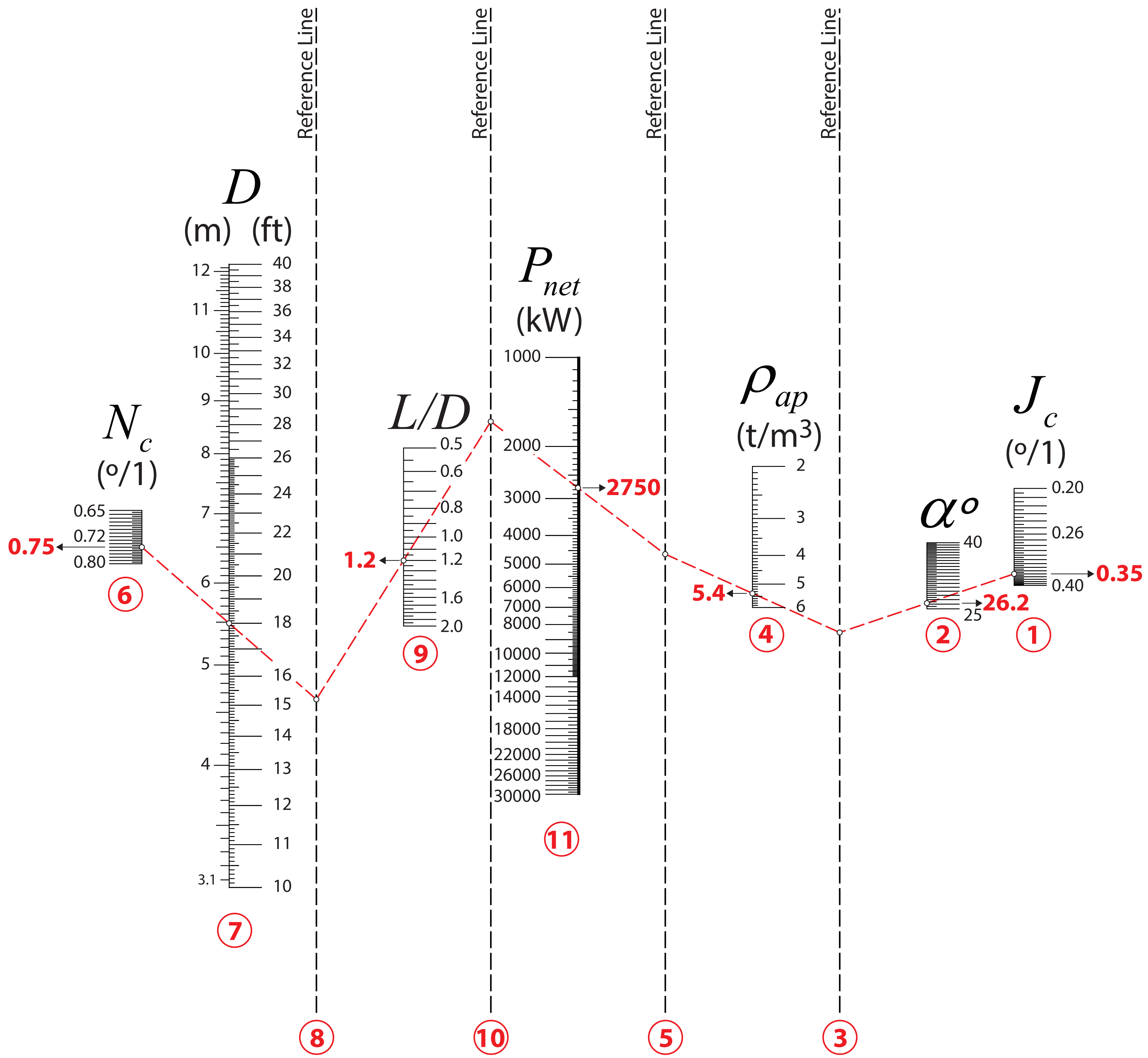
Hogg & Fuerstenau Mill Power Model



$$P_{net} = 0.267 \cdot D^{3.5} \cdot L/D \cdot N_c \cdot \rho_{ap} \cdot \left(J_c - J_c^2\right) \cdot \sin\alpha$$

P_{net} = Net mill power draw, kW

D = Effective mill diameter, ft

L = Effective internal mill length, ft

N_c = Rotational mill speed, expressed as a fraction (°/1) of its critical full entrifugation speed

ρ_{ap} = Apparent mill charge density, t/m^3

J_c = Apparent mill filling, °/1

α = Lift angle of the charge kidney's center of gravity